0.1 Representative Quasi-Newton Update Rules

Given B_k, s_k, and y_k, there exist many update rules that produce B_{k+1} satisfying the secant condition. We introduce several representative ones together with their derivations [3]. In this subsection only, for brevity, we abbreviate B_k, B_{k+1}, s_k, and y_k as B, $\bar{B}$, s, and y, respectively.

0.1.1 Broyden's Update

Broyden's update is one of the oldest known update rules, but it is not often used for quasi-Newton methods in practice due to its lack of symmetry. We introduce it briefly for historical context. The update formulas are given by

$$\bar{B}_{\text{Broyden}} := B + \frac{(y - Bs)s^\top}{s^\top s}.$$

Derivation (Broyden) We derive this formula from simple structural assumptions [3, Section 4].

Proposition 1. *Assume that $\bar{B}$ satisfies the secant condition*

$$\bar{B}s = y,$$

and the following constraint for all vectors $z \in \mathbb{R}^n$ such that $z^\top s = 0$:

$$\bar{B}z = Bz.$$

Then $\bar{B}$ is uniquely determined, and it is $\bar{B}_{\text{Broyden}}$.

Proof. A vector s and a basis for the orthogonal complement of s form a basis of $\mathbb{R}^n$. Since the conditions on $\bar{B}$ completely determine the action of $\bar{B}$ with respect to this basis, $\bar{B}$ is uniquely determined.

We now show that $\bar{B}_{\text{Broyden}}$ satisfies the imposed conditions. Let z be any vector satisfying $z^\top s = 0$. Then

$$\begin{aligned}
\bar{B}_{\text{Broyden}}s &= \left(B + \frac{(y - Bs)s^\top}{s^\top s}\right)s = Bs + (y - Bs) = y,\\
\bar{B}_{\text{Broyden}}z &= \left(B + \frac{(y - Bs)s^\top}{s^\top s}\right)z = Bz + (z^\top s)\frac{y - Bs}{s^\top s} = Bz.
\end{aligned}$$

Hence $\bar{B}_{\text{Broyden}}$ satisfies the imposed conditions.

Therefore, by uniqueness, we conclude that $\bar{B} = \bar{B}_{\text{Broyden}}$. □

Broyden's update can also be characterized as a minimal-change update in the Frobenius norm.

Proposition 2 ([3, Theorem 4.1])**.** *Let $B \in \mathbb{R}^{n\times n}$, $y \in \mathbb{R}^n$, and $s \in \mathbb{R}^n \setminus \{0\}$ be given. Then $\bar{B}_{\text{Broyden}}$ is the unique solution of the following optimization problem:*

$$\begin{aligned}
\underset{\tilde{B}\in\mathbb{R}^{n\times n}}{\text{minimize}} \quad & \left\|\tilde{B} - B\right\|_F\\
\text{subjectto} \quad & \tilde{B}s = y.
\end{aligned}$$

Proof. Instead of directly solving the optimization problem, we note that we can think of optimizing the function $\tilde{B} \mapsto \left\|\tilde{B} - B\right\|_F^2$, which is convex on $\mathbb{R}^{n\times n}$. The constraint set

$$\{\tilde{B} \in \mathbb{R}^{n\times n} : \tilde{B}s = y\}$$

is affine and hence convex. Therefore, the optimization problem admits at most one minimizer.

We show that $\bar{B}_{\text{Broyden}}$ is indeed the minimizer. For any $\tilde{B}$ satisfying $\tilde{B}s = y$, we have

$$\left\|\bar{B}_{\text{Broyden}} - B\right\|_F^2 = \left\|\frac{(y - Bs)s^\top}{s^\top s}\right\|_F^2 \qquad \text{(definition of } \bar{B}_{\text{Broyden}})$$

$$
\begin{aligned}
&= \left\| (\tilde{B} - B) \frac{ss^\top}{s^\top s} \right\|_F^2 && (\tilde{B}s = y) \\
&\leq \left\| \tilde{B} - B \right\|_F^2 .
\end{aligned}
$$

In the last inequality, we used the submultiplicativity of the Frobenius norm and the fact that $\left\| ss^\top/(s^\top s) \right\|_F = 1$. Hence $\tilde{B} = \bar{B}_{\text{Broyden}}$, and we obtain the claim. □

Note that this update rule does not preserve symmetry. In the following, we introduce representative quasi-Newton update rules that improve upon this aspect.

0.1.2 Symmetric Rank-One (SR1) Update

The Symmetric Rank-One (SR1) update [9] is a fundamental quasi-Newton method that maintains symmetry throughout the update process. The update formulas are given by

$$\bar{B}_{\text{SR1}} := B + \frac{(y - Bs)(y - Bs)^\top}{(y - Bs)^\top s}.$$

Derivation (SR1) To derive the SR1 update, we construct the updated matrix $\bar{B}$ as a rank-one update. That is, we assume that for some vector $z \in \mathbb{R}^n$,

$$\bar{B}_{\text{SR1}} = B + zz^\top.$$

To satisfy the secant condition $\bar{B}_{\text{SR1}} s = y$, when $z^\top s \neq 0$, we need

$$Bs + zz^\top s = y.$$

Rearranging this equation gives

$$z = \frac{y - Bs}{z^\top s}.$$

To determine $z^\top s$, we take the inner product with s:

$$z^\top s = \frac{(y - Bs)^\top s}{z^\top s}.$$

Rearranging this equation yields the key relation

$$(z^\top s)^2 = (y - Bs)^\top s.$$

Thus, we obtain

$$\bar{B}_{\text{SR1}} = B + zz^\top = B + \frac{(y - Bs)(y - Bs)^\top}{(z^\top s)^2} = B + \frac{(y - Bs)(y - Bs)^\top}{(y - Bs)^\top s}.$$

Thus, we have derived the SR1 update formula. We omit the derivation of $\bar{H}_{\text{SR1}}$.

Remarks The SR1 update requires $(y - Bs)^\top s \neq 0$ to be well-defined. When $(y - Bs)^\top s = 0$, the denominator of the SR1 update vanishes, and the update is typically skipped in practice.

0.1.3 Powell Symmetric Broyden (PSB) Update

The Powell Symmetric Broyden (PSB) update [2, 4, 8] is also a quasi-Newton update rule. The update formulas are given by

$$\bar{B}_{\text{PSB}} = B + \frac{(y - Bs)s^\top + s(y - Bs)^\top}{s^\top s} - \frac{s^\top (y - Bs)}{(s^\top s)^2} ss^\top$$

Derivation (PSB) Recall that the SR1 update adds $zz^\top$ to B to obtain $\bar{B}$. Instead, we can consider an asymmetric rank-one update of the form $zc^\top$ for some vector $c \in \mathbb{R}^n$. We symmetrize the result at the end. Assume $c^\top s \neq 0$, and from $Bs + zc^\top s = y$, we can derive z as

$$z = \frac{y - Bs}{c^\top s}$$

and perform the asymmetric update

$$C_1 := B + \frac{(y - Bs)c^\top}{c^\top s}.$$

Since C_1 is generally not symmetric, we symmetrize it via

$$C_2 = \frac{C_1 + C_1^\top}{2}.$$

However, the symmetrized matrix C_2 may not satisfy the secant condition $C_2 s = y$. Therefore, we iterate this process:

$$\begin{cases} C_0 = B \\ C_{2t+1} = C_{2t} + \dfrac{(y - C_{2t}s)c^\top}{c^\top s} & \text{(asymmetric update)} \\ C_{2t+2} = \dfrac{C_{2t+1} + C_{2t+1}^\top}{2} & \text{(symmetrization)} \end{cases}$$

The key result is that the sequence $\{C_t\}_t$ converges to a symmetric matrix satisfying the secant condition, as formalized in the following proposition.

Proposition 3 ([3, Lemma 7.2])**.** *Suppose that $c^\top s \neq 0$. Then, the sequence $\{C_t\}_t$ converges, and the limit is given by*

$$\lim_{t \to \infty} C_t = C_\infty := B + \frac{(y - Bs)c^\top + c(y - Bs)^\top}{c^\top s} - \frac{(y - Bs)^\top s}{(c^\top s)^2} cc^\top.$$

Proof. We first analyze the even subsequence. For $t = 0, 1, 2, \ldots$, we define

$$G_t := C_{2t}.$$

By construction, each G_t is symmetric. From the definitions, we have

$$G_{t+1} = G_t + \frac{1}{2c^\top s}\left((y - G_t s)c^\top + c(y - G_t s)^\top\right).$$

Let us introduce the error vector for the secant condition:

$$w_t := y - G_t s.$$

Then, we have

$$G_{t+1} = G_t + \frac{1}{2c^\top s}(w_t c^\top + c w_t^\top).$$

Substituting the above equation into the definition of w_t yields

$$\begin{aligned} w_{t+1} &= y - \left(G_t + \frac{1}{2c^\top s}(w_t c^\top + c w_t^\top)\right) s \\ &= w_t - \frac{1}{2} w_t - \frac{w_t^\top s}{2c^\top s} c \\ &= \frac{1}{2}\left(w_t - \frac{w_t^\top s}{c^\top s} c\right). \end{aligned}$$

Hence, we have

$$w_{t+1} = P w_t, \qquad P := \frac{1}{2}\left(I - \frac{cs^\top}{c^\top s}\right).$$

The matrix $cs^\top/c^\top s$ is rank one and has eigenvalues $1,0,\ldots,0$. Therefore, P has one eigenvalue 0 and all remaining eigenvalues equal to $1/2$. In particular, its spectral radius is $1/2 < 1$, and thus the series converges as follows:

$$\begin{aligned}
\sum_{t=0}^{\infty} w_t &= \sum_{t=0}^{\infty} P^t(y-Bs) && (w_0 = y-Bs)\\
&= (I-P)^{-1}(y-Bs)\\
&= 2\left(I - \frac{1}{2}\frac{cs^\top}{c^\top s}\right)(y-Bs). && (\text{definition of } P)
\end{aligned}$$

Note that the last equation follows from

$$2(I-P)\left(I-\frac{1}{2}\frac{cs^\top}{c^\top s}\right) = \left(I+\frac{cs^\top}{c^\top s}\right)\left(I-\frac{1}{2}\frac{cs^\top}{c^\top s}\right) = I.$$

Hence, we have

$$\begin{aligned}
\lim_{t\to\infty} G_t &= B + \frac{1}{2c^\top s}\sum_{t=0}^{\infty}(w_t c^\top + c w_t^\top)\\
&= B + \left(\sum_{t=0}^{\infty} w_t\right)\frac{c^\top}{2c^\top s} + \frac{c}{2c^\top s}\left(\sum_{t=0}^{\infty} w_t\right)^\top\\
&= B + 2\left(I-\frac{1}{2}\frac{cs^\top}{c^\top s}\right)(y-Bs)\frac{c^\top}{2c^\top s} + \frac{c}{2c^\top s}2(y-Bs)^\top\left(I-\frac{1}{2}\frac{sc^\top}{c^\top s}\right)\\
&= B + \frac{(y-Bs)c^\top + c(y-Bs)^\top}{c^\top s} - \frac{(y-Bs)^\top s}{(c^\top s)^2}cc^\top\\
&= C_\infty.
\end{aligned}$$

This shows the convergence of the even subsequence $\{C_{2t}\}_t$ to C_∞.

Next, for the odd subsequence, we have

$$C_{2t+1} = G_t + \frac{w_t c^\top}{c^\top s}.$$

Since $G_t \to C_\infty$ and $\|w_t\| \to 0$ from the spectral radius of P being less than one, we have

$$C_{2t+1} \to C_\infty.$$

Therefore both subsequences $\{C_{2t}\}_t$ and $\{C_{2t+1}\}_t$ converge to C_∞, completing the proof. $\square$

When $c = s$, the general formula of C_∞ simplifies to the standard PSB update formula:

$$\bar{B}_{\mathrm{PSB}} = B + \frac{(y-Bs)s^\top + s(y-Bs)^\top}{s^\top s} - \frac{(y-Bs)^\top s}{(s^\top s)^2}ss^\top.$$

0.1.4 Davidon–Fletcher–Powell (DFP) Update

The Davidon–Fletcher–Powell (DFP) update [9] is also a classical quasi-Newton update formula. The update formulas are given by

$$\bar{B}_{\mathrm{DFP}} = \left(I - \frac{ys^\top}{y^\top s}\right)B\left(I-\frac{sy^\top}{y^\top s}\right) + \frac{yy^\top}{y^\top s}.$$

Derivation (DFP) In the PSB update rule discussed earlier, we substituted $c = s$, but we can consider taking a different c. Specifically, substituting $c = y$ yields the DFP update as follows.

$$\bar{B}_{\mathrm{DFP}} = B + \frac{(y-Bs)y^\top + y(y-Bs)^\top}{y^\top s} - \frac{(y-Bs)^\top s}{(y^\top s)^2}yy^\top.$$

There are other derivations of the DFP update, some of which can be understood as dual versions of the derivation of the subsequent BFGS update.

0.1.5 Broyden–Fletcher–Goldfarb–Shanno (BFGS) Update

The Broyden–Fletcher–Goldfarb–Shanno (BFGS) update is one of the most widely used quasi-Newton methods. The update formulas are given by

$$\bar{B}_{\mathrm{BFGS}} = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}.$$

Derivation by Duality This update can be derived by considering the dual of the DFP update. The secant condition is $\bar{B}s = y$, and assuming $\bar{B}$ is invertible, we can also write it as $\bar{H}y = s$. By performing the same operations of enforcing the secant condition and symmetrization on this condition, we obtain

$$\bar{H}_{\mathrm{BFGS}} = \left(I - \frac{sy^\top}{y^\top s}\right) H \left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}.$$

As we will see later, this is indeed the inverse of $\bar{B}_{\mathrm{BFGS}}$. Note that this has the same form as the DFP update with (B,s,y) replaced by (H,y,s). Therefore, we can say that the BFGS update is the dual of the DFP update.

Derivation by KL Divergence Minimization The KL divergence is a non-symmetric measure of proximity between matrices. For positive definite matrices $P,Q \in \mathbb{R}^{n\times n}$, it is defined by

$$\mathrm{KL}(P,Q) := \mathrm{tr}(PQ^{-1}) - \log\det\big(PQ^{-1}\big) - n.$$

Let $T \in \mathbb{R}^{n\times n}$ be an arbitrary nonsingular matrix. Since the trace is invariant under cyclic permutations, the KL divergence satisfies

$$\begin{aligned}
\mathrm{KL}(TPT^\top, TQT^\top) &= \mathrm{tr}(TPT^\top(TQT^\top)^{-1}) - \log\det\Big(TPT^\top(TQT^\top)^{-1}\Big) - n \\
&= \mathrm{tr}(PQ^{-1}T^{-1}T) - \log\big(\det(T)\det\big(PQ^{-1}\big)\det\big(T^{-1}\big)\big) - n \\
&= \mathrm{tr}(PQ^{-1}) - \log\det\big(PQ^{-1}\big) - n \\
&= \mathrm{KL}(P,Q).
\end{aligned}$$

It is known that the BFGS update can be obtained as the solution to the following KL divergence minimization problem:

$$\begin{aligned}
\underset{\bar{B}\succ 0}{\text{minimize}} \quad & \mathrm{KL}(\bar{B},B) \\
\text{subject to} \quad & \bar{B}s = y.
\end{aligned}$$

Proposition 4 ([12, Section 7.2.4])**.** *Let $y^\top s > 0$ and $B \succ 0$ be a symmetric positive definite matrix. The solution to the above optimization problem is given by the BFGS update formula.*

Proof. Since B is a symmetric positive definite matrix, we first define

$$s' := B^{\frac{1}{2}}s, \quad y' := B^{-\frac{1}{2}}y, \quad B' := B^{-\frac{1}{2}}\bar{B}B^{-\frac{1}{2}}.$$

Then, the optimization problem can be rewritten as

$$\begin{aligned}
\underset{B'\succ 0}{\text{minimize}} \quad & \mathrm{KL}(B',I) \\
\text{subject to} \quad & B's' = y'.
\end{aligned}$$

Here, we used the invariance of the KL divergence:

$$\mathrm{KL}(\bar{B},B) = \mathrm{KL}(B^{-\frac{1}{2}}\bar{B}B^{-\frac{1}{2}}, B^{-\frac{1}{2}}BB^{-\frac{1}{2}}) = \mathrm{KL}(B',I)$$

Note that the constraint $B' \succ 0$ automatically includes symmetry, so adding the constraint $B' = B'^\top$ explicitly does not change the solution.

Let us solve this problem using the method of Lagrange multipliers. We introduce the Lagrange multipliers $\lambda \in \mathbb{R}^n$ and $\Lambda \in \mathbb{R}^{n\times n}$, and define the Lagrangian as

$$\mathscr{L}(B',\lambda,\Lambda) = \operatorname{tr}(B') - \log\det\left(B'\right) - n + 2\lambda^\top(B's'-y') + \operatorname{tr}\left(\Lambda(B'-B'^\top)\right).$$

In general, for matrix differentiation, the derivative of $\log\det(X)$ with respect to X for $X \succ 0$ is $X^{-\top}$, and the derivative of $\operatorname{tr}(AX)$ is $A^\top$. By differentiating with respect to B' and using $B' = B'^\top$, we can write down the stationarity condition of the KKT conditions as follows:

$$I - B'^{-\top} + 2\lambda s'^\top + \Lambda^\top - \Lambda = 0.$$

By adding this equation and its transpose, and dividing by 2 again using $B' = B'^\top$, we obtain

$$B'^{-1} = I + \lambda s'^\top + s'\lambda^\top.$$

By the constraint $B's' = y'$, we have $B'^{-1}y' = s'$, so multiplying the above equation by y' from the right and then by $y'^\top$ from the left yields

$$\begin{aligned} s' &= y' + (s'^\top y')\lambda + (\lambda^\top y')s', \\ y'^\top s' &= y'^\top y' + (s'^\top y')(y'^\top\lambda) + (\lambda^\top y')(y'^\top s'). \end{aligned}$$

From the second equation, we have

$$\lambda^\top y' = \frac{y'^\top s' - y'^\top y'}{2(s'^\top y')}.$$

Substituting this into the first equation gives

$$\lambda = \frac{s'-y'}{s'^\top y'} - \frac{\lambda^\top y'}{s'^\top y'}s' = \frac{s'^\top y' + y'^\top y'}{2(s'^\top y')^2}s' - \frac{1}{s'^\top y'}y'.$$

By substituting this λ into the expression for B'^{-1} and simplifying, we obtain

$$\begin{aligned} B'^{-1} &= I + 2\frac{s'^\top y' + y'^\top y'}{2(s'^\top y')^2}(s's'^\top) - \frac{1}{s'^\top y'}(y's'^\top + s'y'^\top) \\ &= \left(I - \frac{s'y'^\top}{s'^\top y'}\right)\left(I - \frac{y's'^\top}{s'^\top y'}\right) + \frac{s's'^\top}{s'^\top y'}. \end{aligned}$$

Finally, since $B' = B^{-\frac{1}{2}}\bar{B}B^{-\frac{1}{2}}$, we have $\bar{B} = B^{\frac{1}{2}}B'B^{\frac{1}{2}}$, and thus calculating $\bar{B}^{-1}$ gives

$$\begin{aligned} \bar{B}^{-1} &= B^{-\frac{1}{2}}B'^{-1}B^{-\frac{1}{2}} \\ &= B^{-\frac{1}{2}}\left(I - \frac{s'y'^\top}{s'^\top y'}\right)\left(I - \frac{y's'^\top}{s'^\top y'}\right)B^{-\frac{1}{2}} + \frac{(B^{-\frac{1}{2}}s')(B^{-\frac{1}{2}}s')^\top}{s'^\top y'} \\ &= B^{-\frac{1}{2}}\left(I - \frac{B^{\frac{1}{2}}sy^\top B^{-\frac{1}{2}}}{s^\top y}\right)\left(I - \frac{B^{-\frac{1}{2}}ys^\top B^{\frac{1}{2}}}{s^\top y}\right)B^{-\frac{1}{2}} + \frac{ss^\top}{s^\top y} \\ &= \left(I - \frac{sy^\top}{y^\top s}\right)B^{-1}\left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}. \end{aligned}$$

As we will see later, this is indeed the inverse of the BFGS update and positive definite, the proof is complete. □

As for more details, see [5, 6].

0.2 Details of BFGS Update

In this subsection, we focus on the BFGS update and provide a detailed derivation of its formula. BFGS update is known as one of the most successful quasi-Newton update rules in practice. Recall that the BFGS update formula is given by

$$B_{k+1} = B_k - \frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k}.$$

0.2.1 Formula for the Inverse Update

The inverse of the BFGS update $H_k := B_k^{-1}$ is given by

$$H_{k+1} = \left(I - \frac{s_k y_k^\top}{y_k^\top s_k}\right) H_k \left(I - \frac{y_k s_k^\top}{y_k^\top s_k}\right) + \frac{s_k s_k^\top}{y_k^\top s_k}.$$

Note that this BFGS update formula for H_{k+1} has the same form as the DFP update formula for B_{k+1}. These are dual to each other. We now prove that this equation indeed gives the inverse of the BFGS update.

Proposition 5. *The matrix H_{k+1} is the inverse of B_{k+1}.*

Proof. The BFGS update can be rewritten in a compact rank-two form

$$B_{k+1} = B_k + UCV^\top.$$

Here, we define

$$U = \begin{bmatrix} B_k s_k & y_k \end{bmatrix}, \qquad C = \begin{pmatrix} -\frac{1}{s_k^\top B_k s_k} & 0 \\ 0 & \frac{1}{y_k^\top s_k} \end{pmatrix}, \qquad V = \begin{bmatrix} B_k s_k & y_k \end{bmatrix}$$

since

$$UCV^\top = \begin{bmatrix} -\frac{B_k s_k}{s_k^\top B_k s_k} & \frac{y_k}{y_k^\top s_k} \end{bmatrix} \begin{bmatrix} s_k^\top B_k \\ y_k^\top \end{bmatrix} = -\frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k}.$$

By the Sherman–Morrison–Woodbury identity, we can compute H_{k+1} as follows:

$$\begin{aligned}
H_{k+1} &= (B_k + UCV^\top)^{-1} \\
&= B_k^{-1} - B_k^{-1} U \left(C^{-1} + V^\top B_k^{-1} U\right)^{-1} V^\top B_k^{-1} \\
&= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \left(\begin{pmatrix} -s_k^\top B_k s_k & 0 \\ 0 & y_k^\top s_k \end{pmatrix} + \begin{pmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top H_k y_k \end{pmatrix} \right)^{-1} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\
&= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \begin{pmatrix} 0 & y_k^\top s_k \\ y_k^\top s_k & y_k^\top H_k y_k + y_k^\top s_k \end{pmatrix}^{-1} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\
&= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \left(-\frac{1}{(y_k^\top s_k)^2} \begin{pmatrix} y_k^\top H_k y_k + y_k^\top s_k & -y_k^\top s_k \\ -y_k^\top s_k & 0 \end{pmatrix} \right) \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\
&= H_k + \frac{1}{(y_k^\top s_k)^2} \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \begin{pmatrix} y_k^\top H_k y_k + y_k^\top s_k & -y_k^\top s_k \\ -y_k^\top s_k & 0 \end{pmatrix} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\
&= H_k + \frac{1}{(y_k^\top s_k)^2} \left((y_k^\top H_k y_k + y_k^\top s_k) s_k s_k^\top - (y_k^\top s_k)(s_k y_k^\top H_k + H_k y_k s_k^\top) \right) \\
&= \left(I - \frac{s_k y_k^\top}{y_k^\top s_k}\right) H_k \left(I - \frac{y_k s_k^\top}{y_k^\top s_k}\right) + \frac{s_k s_k^\top}{y_k^\top s_k}.
\end{aligned}$$

which completes the proof. □

0.2.2 Positive Definiteness of the BFGS Update

An important property of the BFGS update is that if the current approximation B_k is positive definite and the curvature condition $y_k^\top s_k > 0$ holds, then the updated approximation B_{k+1} is also guaranteed to be positive definite.

Proposition 6. *If B_k is positive definite and $y_k^\top s_k > 0$, then B_{k+1} is also positive definite.*

Proof. By assumption, B_k and its inverse H_k are positive definite. For any nonzero vector $v \in \mathbb{R}^n$, we have

$$v^\top H_{k+1} v = v^\top \left(I - \frac{s_k y_k^\top}{y_k^\top s_k}\right) H_k \left(I - \frac{y_k s_k^\top}{y_k^\top s_k}\right) v + v^\top \frac{s_k s_k^\top}{y_k^\top s_k} v \geq 0 + \frac{(s_k^\top v)^2}{y_k^\top s_k} > 0.$$

The first term is nonnegative since H_k is positive definite, and the second term is positive due to the curvature condition $y_k^\top s_k > 0$. Thus, H_{k+1} is positive definite, and consequently, $B_{k+1} = H_{k+1}^{-1}$ is also positive definite. □

This proposition is crucial for ensuring the positive definiteness of the approximate Hessian matrices B_k throughout the iterations, which in turn guarantees that the search directions are descent directions. In practice, it is common to use a line search that satisfies the following strong curvature condition, included in the strong Wolfe conditions:

$$|g_{k+1}^\top d_k| \leq c_2 |g_k^\top d_k|.$$

Here, $0 < c_2 < 1$ is a constant. By imposing this condition, we can verify that B_k remains positive definite throughout the iterations. First, we initialize B_0 to be positive definite. Assuming that B_k is positive definite for some k, since d_k is a descent direction with $g_k^\top d_k < 0$, and from $s_k = \alpha_k d_k$ and $y_k = g_{k+1} - g_k$, we have

$$\begin{aligned}
y_k^\top s_k &= (g_{k+1} - g_k)^\top (\alpha_k d_k) \\
&= \alpha_k (g_{k+1}^\top d_k - g_k^\top d_k) \\
&\geq \alpha_k \left(-|g_{k+1}^\top d_k| + |g_k^\top d_k| \right) && (g_k^\top d_k < 0) \\
&\geq \alpha_k (1 - c_2) |g_k^\top d_k| && \text{(by the curvature condition)} \\
&> 0. && (c_2 < 1)
\end{aligned}$$

Therefore, the curvature condition $y_k^\top s_k > 0$ is satisfied, and by the previous proposition, B_{k+1} is also positive definite. Thus, by mathematical induction, all B_k are guaranteed to be positive definite for every k.

0.2.3 Trace and Determinant Formulas for the BFGS Update

We also present trace and determinant formulas for the BFGS update, which are useful in analyzing the eigenvalue behavior of the updated matrix. For symmetric matrices, these quantities correspond to the sum and product of eigenvalues, respectively. Hence, if both the trace and the determinant are appropriately bounded, one may expect the eigenvalues themselves to remain bounded. This is closely related to assumptions on the Hessian eigenvalues of the objective function, such as μ-strong convexity and L-smoothness.

Trace Formula The trace of the BFGS-updated matrix has an explicit formula.

Proposition 7 ([9, eq. 6.44])**.** *Let $B_+ = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}$ be the BFGS update. Then*

$$\operatorname{tr}(B_+) = \operatorname{tr}(B) - \frac{\|Bs\|^2}{s^\top Bs} + \frac{\|y\|^2}{y^\top s}.$$

Proof. Applying the trace to the BFGS update formula, we have

$$\operatorname{tr}(B_+) = \operatorname{tr}(B) - \operatorname{tr}\left(\frac{Bss^\top B}{s^\top Bs}\right) + \operatorname{tr}\left(\frac{yy^\top}{y^\top s}\right).$$

For the second term, note that

$$\operatorname{tr}\left(\frac{Bss^\top B}{s^\top Bs}\right) = \frac{1}{s^\top Bs}\operatorname{tr}((Bs)(Bs)^\top) = \frac{\|Bs\|^2}{s^\top Bs}.$$

For the third term, we have

$$\operatorname{tr}\left(\frac{yy^\top}{y^\top s}\right) = \frac{1}{y^\top s}\operatorname{tr}(yy^\top) = \frac{\|y\|^2}{y^\top s}.$$

Combining these results yields the desired formula. □

Determinant Formula The determinant of the BFGS-updated matrix also admits a closed-form expression.

Proposition 8 ([9, eq. 6.45])**.** *Let $B_+ = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}$ be the BFGS update and assume that B is nonsingular. Then*

$$\det(B_+) = \det(B)\frac{y^\top s}{s^\top Bs}.$$

Proof. Recall the rank-two formulation of the BFGS update. By the matrix determinant lemma, the matrices U, C, V satisfy

$$\det(B_{k+1}) = \det\left(B_k + UCV^\top\right) = \det(B_k)\det(C)\det\left(C^{-1} + V^\top B_k^{-1} U\right).$$

Since $U = V = \begin{bmatrix} B_k s_k & y_k \end{bmatrix}$, we have

$$V^\top B_k^{-1} U = \begin{bmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top B_k^{-1} y_k \end{bmatrix},$$

and thus

$$C^{-1} + V^\top B_k^{-1} U = \begin{bmatrix} -s_k^\top B_k s_k & 0 \\ 0 & y_k^\top s_k \end{bmatrix} + \begin{bmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top B_k^{-1} y_k \end{bmatrix} = \begin{bmatrix} 0 & s_k^\top y_k \\ y_k^\top s_k & y_k^\top s_k + y_k^\top B_k^{-1} y_k \end{bmatrix}.$$

Thus, combining these results, we obtain

$$\det(B_{k+1}) = \det(B_k)\left(-\frac{1}{s_k^\top B_k s_k} \cdot \frac{1}{y_k^\top s_k}\right)\left(-(s_k^\top y_k)(y_k^\top s_k)\right) = \det(B_k)\frac{y_k^\top s_k}{s_k^\top B_k s_k},$$

which completes the proof. □

0.3 BFGS vs DFP

As mentioned several times earlier, although BFGS and DFP are dual to each other, they exhibit remarkably different practical efficiency when applied to real optimization problems. Powell's analysis [10] investigates this asymmetry by studying the behavior of both methods on a simple two-dimensional quadratic function. While asymptotic convergence theory often suggests that both methods should perform similarly, Powell reveals that their practical efficiency can differ dramatically, especially when the approximate Hessian is far from the true Hessian. Here, we reproduce his numerical experiments.

0.3.1 Problem Setup

Following Powell's framework, we consider the quadratic function

$$f(x,y) = \frac{1}{2}(x^2 + y^2).$$

The key aspect of this experiment is the choice of the initial Hessian approximation B_0. In principle, the true Hessian of this function is the identity matrix, but here we deliberately use a significantly incorrect initial approximation to test the robustness and corrective ability of the methods against such errors. Specifically, we choose the initial Hessian approximation B_0 to have eigenvalues 1 and λ_1, where λ_1 represents the degree of error in the initial approximation. Both BFGS and DFP methods used a fixed step size of $\alpha_k = 1$ at every iteration.

Additionally, we chose the initial point x_0 as follows:

$$\theta = \arctan\left(\sqrt{\lambda_1}\right), \quad x_0 = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}.$$

This aligns with Powell's original analysis. See [10] for details on this choice.

The iteration continues until the norm of the current point falls below a tolerance relative to its initial norm. For each value of λ_1, we record the number of iterations required for convergence.

0.3.2 Experimental Results

The numerical results are presented in Table 1, which partially reproduces the content of Powell's original tables. We can see that the convergence behavior depends critically on the initial eigenvalue λ_1.

Figure 1 and Figure 2 illustrate the iterative trajectories for specific values of λ_1. These figures visualize how the two methods navigate toward the minimum (at the origin) from the same initial point, clearly showing the difference in convergence speed and path.

λ_1	BFGS	DFP
0.001	4	3
0.01	5	3
0.1	6	4
1	1	1
10	8	16
100	10	107
1000	12	1006
10000	15	9987

Table 1: Convergence comparison between BFGS and DFP methods for different initial eigenvalues λ_1.

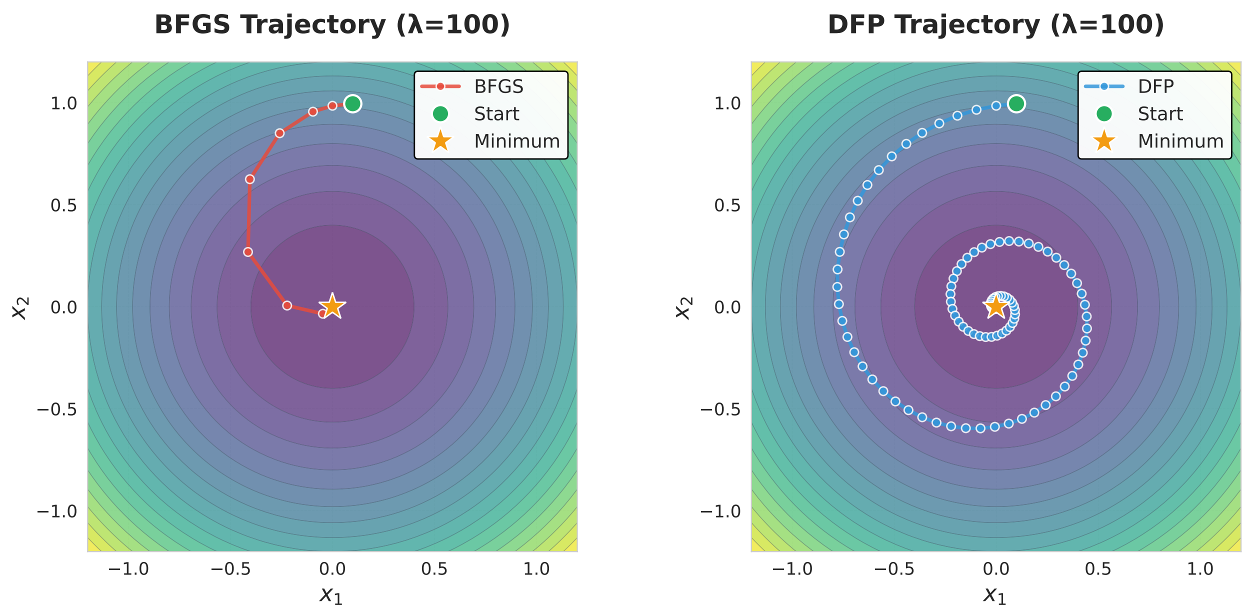

Figure 1: Iterative trajectories for BFGS and DFP methods when $\lambda_1 = 100$. BFGS converges in 10 iterations while DFP requires 107 iterations, showing BFGS's superior efficiency for large eigenvalue errors.

0.3.3 Analysis and Discussion

The numerical results reveal a striking asymmetry between BFGS and DFP. When $\lambda_1 > 1$ (i.e., the initial Hessian approximation overestimates the true Hessian curvature), BFGS demonstrates significantly better efficiency. Conversely, when $\lambda_1 < 1$ (i.e., the initial Hessian approximation underestimates the true curvature), DFP performs slightly better than BFGS, though the difference is modest. This trend reversal can be predicted by the theoretical symmetry between the two methods, but the asymmetry in the magnitude is noteworthy.

Asymmetry in Hessian Correction The asymmetry of the performance arises from a fundamental difference in how these two methods correct erroneous eigenvalues. The core insight is that correcting erroneously large eigenvalues is more critical than correcting erroneously small ones.

When a Hessian eigenvalue is overestimated, the algorithm takes steps that are too conservative, resulting in slow progress of the optimization. Correcting such errors requires the update formula to reduce these large eigenvalues toward one. The BFGS update is highly effective at this task, while the DFP update struggles.

In contrast, when a Hessian eigenvalue is underestimated, the algorithm takes steps that are slightly too aggressive, but the error is self-correcting. Subsequent gradient computations provide information that helps refine the approximation. Thus, correcting underestimated eigenvalues is inherently easier and requires fewer iterations.

This explains why the asymmetry in performance exists between BFGS and DFP.

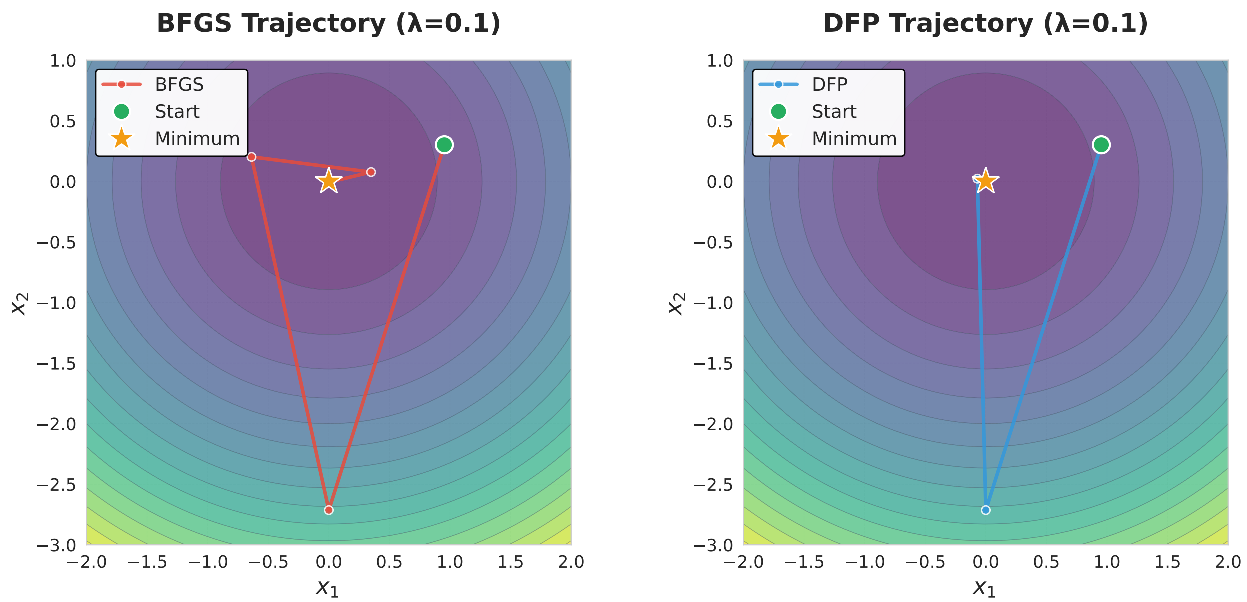

Figure 2: Iterative trajectories for BFGS and DFP methods when $\lambda_1 = 0.1$. Both methods converge quickly, with DFP slightly faster than BFGS.

0.4 Limited Memory BFGS (L-BFGS)

Next, we discuss the limited memory quasi-Newton methods, which are crucial for extending quasi-Newton methods to large-scale optimization problems. In standard quasi-Newton methods, the approximate Hessian B_k or its inverse H_k is stored and updated explicitly as a dense matrix, requiring $\mathcal{O}(n^2)$ memory for n variables. In contrast, limited memory quasi-Newton methods maintain only a limited amount of information, i.e., the most recent m curvature pairs $\{(s_i, y_i)\}$ and perform computations related to the approximate Hessian using just this information. This reduces the space complexity to $\mathcal{O}(nm)$, which is a dramatic improvement when m is a small constant (typically $m \leq 10$).

The L-BFGS method [7], which is a limited memory version of the BFGS update, is particularly representative. We focus on this BFGS update case and show how the quasi-Newton direction $d_k = -H_k g_k$ can be computed efficiently in terms of both space and time complexity using only the limited information.

Throughout this subsection, we work with a finite sequence of matrices.

$$H_0, H_1, \ldots, H_m.$$

Here, H_ℓ denotes the inverse Hessian approximation obtained after ℓ BFGS updates applied to a given initial matrix H_0. This is done for the sake of clarity in the discussion, but it can also be thought of as maintaining a rolling window of the most recent m inverse Hessian approximations corresponding to the past m BFGS updates when the current iteration count is k.

0.4.1 Compact Representation of the Inverse Hessian

Let $\{(s_i, y_i)\}_{i=0}^{m-1}$ be the stored correction pairs, and define

$$\rho_i = \frac{1}{y_i^\top s_i}, \qquad V_i = I - \rho_i y_i s_i^\top.$$

Then, for $i = 0, \ldots, m-1$, the inverse BFGS update can be expressed as

$$H_{i+1} = V_i^\top H_i V_i + \rho_i s_i s_i^\top.$$

By recursively expanding this relation, we obtain the compact representation

$$H_m = V_{m-1}^\top \cdots V_0^\top H_0 V_0 \cdots V_{m-1} + \left(\sum_{j=0}^{m-2} (V_{m-1}^\top \cdots V_{j+1}^\top) \rho_j s_j s_j^\top (V_{j+1} \cdots V_{m-1}) \right) + \rho_{m-1} s_{m-1} s_{m-1}^\top.$$

H_0 denotes the chosen initial inverse Hessian approximation, typically a scaled identity matrix.

0.4.2 Two-Loop Recursion

In quasi-Newton methods, it is crucial for the algorithm to efficiently compute $r = H_m q$ for the inverse matrix H_m and a given vector $q \in \mathbb{R}^n$. This operation can be carried out using two short loops of length m, known as the L-BFGS two-loop recursion [9, Algorithm 7.4]. This algorithm requires $\mathcal{O}(nm)$ time and space complexity.

Algorithm 1: L-BFGS Two-Loop Recursion for $r = H_m q$

Input: q, stored pairs $\{(s_i, y_i)\}_{i=0}^{m-1}$, initial matrix H_0
Output: $r = H_m q$
for $i = m-1, m-2, \ldots, 0$ **do**
    $\rho_i \leftarrow 1/(y_i^\top s_i)$
    $\alpha_i \leftarrow \rho_i s_i^\top q$
    $q \leftarrow q - \alpha_i y_i$
end
$r \leftarrow H_0 q$
for $i = 0, \ldots, m-1$ **do**
    $\beta_i \leftarrow \rho_i y_i^\top r$
    $r \leftarrow r + s_i(\alpha_i - \beta_i)$
end
return r

In the following, we verify that the output of this two-loop recursion algorithm indeed computes $r = H_m q$.

Proposition 9. *The output of the two-loop recursion algorithm satisfies $r = H_m q$.*

Proof. In the first loop of the algorithm ($i = m-1, m-2, \ldots, 0$), from the input vector $q^{(m)} := q$, the algorithm computes

$$\alpha_i := \rho_i s_i^\top q^{(i+1)}, \qquad q^{(i)} := q^{(i+1)} - \alpha_i y_i.$$

Substituting the definition of α_i and using the definition of V_i in the compact representation, we find

$$q^{(i)} = q^{(i+1)} - \rho_i \left(s_i^\top q^{(i+1)} \right) y_i = \left(I - \rho_i y_i s_i^\top \right) q^{(i+1)} = V_i q^{(i+1)}.$$

Thus, for all $i = 0, 1, \ldots, m-1$, we have

$$q^{(i)} = V_i V_{i+1} \cdots V_{m-1} q.$$

Then, the algorithm applies H_0:

$$r^{(0)} = H_0 q^{(0)} = H_0 V_0 V_1 \cdots V_{m-1} q.$$

Next, for $i = 0, 1, \ldots, m-1$, the second loop computes

$$\beta_i = \rho_i y_i^\top r^{(i)}, \qquad r^{(i+1)} = r^{(i)} + s_i \left(\alpha_i - \beta_i \right).$$

Substituting the definitions of α_i, β_i, and $q^{(i+1)}$, we find that when $i < m-1$,

$$\begin{aligned} r^{(i+1)} &= r^{(i)} + s_i \left(\rho_i s_i^\top q^{(i+1)} - \rho_i y_i^\top r^{(i)} \right) \\ &= r^{(i)} + \rho_i s_i s_i^\top \left(V_{i+1} V_{i+2} \cdots V_{m-1} \right) q - \rho_i s_i y_i^\top r^{(i)} \\ &= \left(I - \rho_i y_i s_i^\top \right) r^{(i)} + \rho_i s_i s_i^\top \left(V_{i+1} V_{i+2} \cdots V_{m-1} \right) q \\ &= V_i^\top r^{(i)} + \rho_i s_i s_i^\top \left(V_{i+1} V_{i+2} \cdots V_{m-1} \right) q, \end{aligned}$$

and when $i = m-1$,

$$r^{(m)} = V_{m-1}^\top r^{(m-1)} + \rho_{m-1} s_{m-1} s_{m-1}^\top q.$$

By recursively expanding this relation from the initial value $r^{(0)} = H_0 q^{(0)}$, we obtain

$$r^{(m)} = V_{m-1}^\top \cdots V_0^\top H_0 V_0 \cdots V_{m-1} q + \left(\sum_{j=0}^{m-2} (V_{m-1}^\top \cdots V_{j+1}^\top) \rho_j s_j s_j^\top (V_{j+1} \cdots V_{m-1}) q \right) + \rho_{m-1} s_{m-1} s_{m-1}^\top q.$$

This matches the compact representation of H_m applied to q, and thus we have shown that the output of the algorithm satisfies $r = H_m q$, completing the proof. □

Thus, the two-loop recursion exactly evaluates the action of the matrix H_m on a vector without explicitly constructing the matrix H_m.

0.4.3 Initial Step Size

A crucial component of the L-BFGS method is the choice of the initial matrix H_0. A widely used option is a scaled identity:

$$H_0 = \gamma I.$$

Here, the scaling parameter γ is often chosen as

$$\gamma = \frac{s_{m-1}^\top y_{m-1}}{y_{m-1}^\top y_{m-1}}.$$

Note that the pair (s_{m-1}, y_{m-1}) is the most recent one. This choice is motivated by the following discussion [7, 11]. Assume that the objective function f is twice continuously differentiable and consider the mean Hessian along the most recent step:

$$\bar{G} = \int_0^1 \nabla^2 f(x + \tau s_{m-1}) \mathrm{d}\tau.$$

Recall that s_{m-1} denotes the most recent displacement of x, satisfying

$$y_{m-1} = \nabla f(x + s_{m-1}) - \nabla f(x) = \bar{G} s_{m-1}.$$

Using this relation, the scaling factor can be written as

$$\frac{s_{m-1}^\top y_{m-1}}{y_{m-1}^\top y_{m-1}} = \frac{(\bar{G}^{1/2} s_{m-1})^\top (\bar{G}^{1/2} s_{m-1})}{(\bar{G}^{1/2} s_{m-1})^\top \bar{G} (\bar{G}^{1/2} s_{m-1})}.$$

This is a reciprocal of a Rayleigh quotient of the matrix $\bar{G}$ with respect to the vector $\bar{G}^{1/2} s_{m-1}$. Therefore, this scaling roughly approximates the average inverse curvature along those directions.

Moreover, this choice of γ coincides with the short step size of the Barzilai–Borwein method [1], highlighting a close connection between L-BFGS initialization and classical step-length selection strategies.

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